Molecular Dynamics

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In this project, molecular dynamics was studied through simulations of star interactions using the Verlet method.

I. SIMULATION OF THE INTERACTION BETWEEN TWO STARS

A cluster is a gravitationally bound stellar system. In the absence of external forces, its structure is defined by the balance between gravitational interaction, which tends to bring stars together, and the random distribution of velocities, which tends to disperse them. The dynamics of the interaction between two stars and their orbit was first studied. To this end, it was assumed that stars are material particles, restricting this study to gravitational forces. To simulate the positions $\vec{r}$, such that $r_i = \sqrt{x_i^2 + y_i^2}$, the Verlet Method was implemented for each particle $i \in [1; 2]$, which is a reversible algorithm and does not depend on velocity. Additionally, it has a truncation error of order Δt^4, so attention must be paid to the time step size (it cannot be too small, otherwise it diverges).

$$\vec{r_i}(t + \Delta t) = 2\vec{r_i}(t) - \vec{r_i}(t - \Delta t) + \vec{a_i}(t)\Delta t^2 \tag{1}$$

To apply Verlet, since two positions $\vec{r}(t)$ and $\vec{r}(t - \Delta t)$ are required, an Euler step is used to obtain $\vec{r}(t - \Delta t)$

$$\vec{r_i}(t - \Delta t) = \vec{r_i}(t) - \vec{v_i}(t)\Delta t \tag{2}$$

Since the potential is central, the trajectories of the stars can be projected onto a plane. The accelerations caused by the gravitational force exerted by the stars are calculated, for each star, by

$$\vec{a_i} = -G \sum_{i \neq j}^{N} \frac{m_j}{\vec{r_{ij}}} \tag{3}$$

where $\vec{r_{ij}}$ represents the vector describing the distance between the two interacting stars, and G is the gravitational constant. Next, the velocities were calculated using central differences:

$$\vec{v_i} = \frac{\vec{r_i}(t + \Delta t) - \vec{r_i}(t - \Delta t)}{2\Delta t} \tag{4}$$

Thus, with these data, it is possible to study the energetic considerations of this interaction.

Observing Figure 1, it is noted that E_m is constant (as expected, since the system is conservative) and negative, due to the trajectories being closed elliptical orbits with eccentricity less than 1. The peaks of E_c and E_p occur at the point of maximum proximity between the two stars. $\epsilon_{E_m} = 4.2556e - 8\%$ was calculated, a value resulting from the truncation errors of Verlet, but still highly satisfactory.

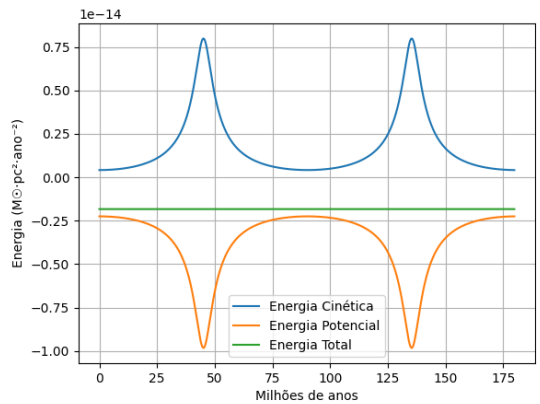

Figure 1: Temporal evolution of kinetic, potential, and mechanical energy of a binary system composed of two solar-mass stars. The initial positions are $(x_1, y_1) = (-1, 0)$ pc and $(x_2, y_2) = (1, 0)$ pc, with initial velocities $(v_{x,1}, v_{y,1}) = (0, 0.02)$ km/s and $(v_{x,2}, v_{y,2}) = (0, -0.02)$ km/s. Integration was performed using the Verlet method, with a time step of $\Delta t = 10^5$ years.

II. SIMULATION OF STELLAR CLUSTER DYNAMICS

The objective now is to extend the previously studied interaction to multiple stars instead of just two. To achieve this, files containing initial positions, initial velocities, and masses of clusters with $N = \{10, 100, 500\}$ stars were used. Interactions were studied using the Verlet method, as before, and at each time step the coordinates of the center of mass were computed, as well as the distances of stars to it, counting how many stars were at a distance smaller than $rt = 5pc$ from the cluster's center of mass. Figure 2 shows that the cluster with 500 stars has the highest number of stars confined within 5 pc. This behavior results from the fact that a more populated cluster has greater total mass, generating a deeper gravitational potential. Consequently, the escape veloc-

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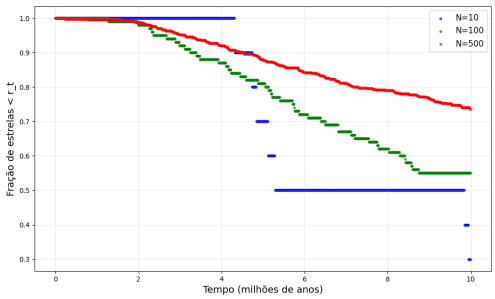

Figure 2: Fraction of stars with distance to the center of mass d smaller than $r_t = 5\,\mathrm{pc}$ as a function of time. The time step used was $\Delta t = 10$ thousand years.

ity increases, making it less likely for stars to gain enough energy to leave the system. This leads to greater stellar retention within the considered radius.

III. III. GENERATION OF THE INITIAL CONFIGURATION OF THE STELLAR CLUSTER FOR DYNAMICS SIMULATION

The goal of this exercise is to construct different initial configurations of a cluster containing 100 stars, each characterized by mass, position, and three-dimensional velocity. Repeating this process generates a statistically large sample, allowing computation of the mean and standard deviation of the curve shown in Figure 3, resulting from the dynamical evolution of 1000 distinct but statistically equivalent clusters. The initial mass distribution is determined by the Kroupa initial mass function, given by:

$$\xi(m) = \begin{cases} m^{-0.3}, & m < 0.08\,M_\odot, \\ m^{-1.3}, & 0.08 \le m < 0.5\,M_\odot, \\ m^{-2.3}, & m > 0.5\,M_\odot, \end{cases} \tag{4}$$

with m expressed in solar units. To generate masses numerically, the cumulative function is constructed and a uniform random number $\xi \in [0,1]$ is generated; the corresponding mass interval is identified and the appropriate CDF transformation is applied. The spatial distribution of stars is described by the King profile:

$$n(r) = n_0 \left[\left(1 + \frac{r^2}{r_c^2}\right)^{-1/2} - \left(1 + \frac{r_t^2}{r_c^2}\right)^{-1/2} \right]^2, \qquad r < r_t, \tag{5}$$

where $r_c = 1\,\mathrm{pc}$ and $r_t = 5\,\mathrm{pc}$. Since the CDF of this profile does not admit analytical inversion, the rejection method was used: generate $r \in [0, r_t]$, compute $n(r)$, and compare with the maximum value $n(0)$; accept if $u < n(r)/n(0)$, with u uniformly generated in $[0,1]$. The spatial orientations of positions and velocities are uniformly distributed on the sphere using two independent random variables $u, v \in [0,1]$:

$$\theta = 2\pi u, \qquad \phi = \cos^{-1}(2v - 1), \tag{7}$$

with Cartesian coordinates given by:

$$x = r \sin\phi \cos\theta, \quad y = r \sin\phi \sin\theta, \quad z = r \cos\phi.$$

The velocity magnitude is obtained from the Maxwell–Boltzmann distribution:

$$f(v)\, d^3v = (\pi v_p^2)^{-3/2} \exp\left(-\frac{v^2}{v_p^2}\right) d^3v, \tag{6}$$

with $v_p = 1\,\mathrm{km/s}$. The Cartesian components are generated using the Box–Muller method, producing normally distributed variables; the direction is assigned as in the case of positions, ensuring isotropy. Repeating all these procedures 1000 times yields 1000 physically consistent initial configurations. These are used as initial conditions in dynamical simulations, allowing computation of the mean curve and standard deviation of the fraction of stars within the tidal radius over time, as shown in Figure 3.

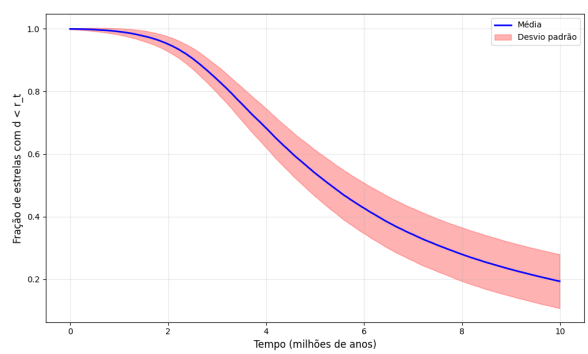

Figure 3: Mean (blue) fraction of stars with distance to the center of mass d smaller than $r_t = 5\,\mathrm{pc}$, obtained from 1000 independent realizations, as a function of time. The simulation was performed using the Verlet method with a time step of $\Delta t = 10$ thousand years. The error bars (red) represent the standard deviation computed at each time step.